THE PROPERTY THAT A MORPHISM INDUCES AN ISOMORPHISM OF AUTOMORPHISM GROUPS IS INDEPENDENT OF THE CHOSEN POINT IN ITS CLASS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The automorphism group algebraic spaces of a point of an algebraic stack, and the equivalence relation on such points given by field extensions. The author's section on automorphism groups, in which the counted statement occurs. Fixes the two maps whose being isomorphisms is asserted to be equivalent.
- The descent of the property of being an isomorphism along a faithfully flat and quasi-compact morphism. Invoked at the start of the counted proof (source lines 4030-4033) as a cross-chapter pointer. Shows that the property propagates along a field extension, whence along the equivalence class of the point.

Source reference. Original reference spaces-descent-lemma-descending-property-isomorphism: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 4033–4033. Navigation pointer only: the counted text refers to the Stacks item spaces-descent-lemma-descending-property-isomorphism, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Let $x \in |\mathcal{X}|$ be a point. The following are equivalent*

(1) *for some morphism $x : \mathrm{Spec}(k) \to \mathcal{X}$ in the class of x setting $y = f \circ x$ the map $G_x \to G_y$ of automorphism group algebraic spaces is an isomorphism, and*

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(2) *for any morphism* $x : \mathrm{Spec}(k) \to \mathcal{X}$ *in the class of* x *setting* $y = f \circ x$ *the map* $G_x \to G_y$ *of automorphism group algebraic spaces is an isomorphism.*

Proof. This comes down to the fact that being an isomorphism is fpqc local on the target, see Descent on Spaces, Lemma spaces-descent-lemma-descending-property-isomorphism. Namely, suppose that k'/k is an extension of fields and denote $x' : \mathrm{Spec}(k') \to \mathcal{X}$ the composition and set $y' = f \circ x'$. Then the morphism $G_{x'} \to G_{y'}$ is the base change of $G_x \to G_y$ by $\mathrm{Spec}(k') \to \mathrm{Spec}(k)$. Hence $G_x \to G_y$ is an isomorphism if and only if $G_{x'} \to G_{y'}$ is an isomorphism. Thus we see that the property propagates through the equivalence class if it holds for one. □